

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (CSE) (Credit System) Degree Examinations

January 2023

19CSE13 – OPERATIONS RESEARCH

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Normalization table is permitted.

Unit – I

- | | Marks | BT* | CO* | PO* | | | | | | | | | | | | | | | | | | |
|---|--------------|--------------------------------------|-----|------|--------------------|--------------------|---|---|---|---|---|---|------|---|---|---|---|------|--|--|--|--|
| 1. a) List the various applications of Operations Research. | 4 | L*2 | 1 | 1 | | | | | | | | | | | | | | | | | | |
| b) Solve the following LPP using the graphical method
Max $Z=3X_1+2X_2$ with constraints
$-2X_1+X_2 \leq 1$
$X_1 \leq 2$
$X_1+X_2 \leq 3$ $X_1, X_2 \geq 0$ | 8 | L3 | 2 | 1 | | | | | | | | | | | | | | | | | | |
| c) Find the dual of the following primal LPP
Min $Z=2X_1+3X_2+4X_3$ with constraints
$2X_1+3X_2+5X_3 \geq 2$
$3X_1+X_2+7X_3 = 3$
$X_1+4X_2+6X_3 \leq 5$ $X_1 \geq 0$, X_2 and X_3 are unrestricted | 8 | L3 | 2 | 1 | | | | | | | | | | | | | | | | | | |
| 2. a) Solve the following LPP using Dual Simplex method.
Min $Z=6X_1+7X_2+3X_3+5X_4$ with constraints
$5X_1+6X_2-3X_3+4X_4 \geq 12$
$2X_1+5X_2+X_3+X_4 \geq 8$
$X_2+5X_3-6X_4 \geq 10$ $X_1, X_2, X_3, X_4 \geq 0$ | 8 | L3 | 2 | 1 | | | | | | | | | | | | | | | | | | |
| b) Solve the following LPP using Revised Simplex method.
Max $Z=3X_1+5X_2$ with constraints
$3X_1+2X_2 \leq 18$
$X_1 \leq 4$
$X_2 \leq 6$ $X_1, X_2 \geq 0$ | 12 | L4 | 2 | 1 | | | | | | | | | | | | | | | | | | |
| 3. a) A firm manufactures 3 products A, B and C. Time to manufacture product A is twice that for B and thrice that for C and if the entire labor is engaged in making product A, 1600 units of this product can be produced. These products are to be produced in the ratio 3:4:5. There is demand for at least 300, 250 and 200 units of products A, B and C and the profit earned per unit is Rs. 90, 40 and 30 respectively. The raw material requirement is as shown below: | | | | | | | | | | | | | | | | | | | | | | |
| <table border="1"> <thead> <tr> <th rowspan="2">Raw material</th> <th colspan="3">Requirement for unit of product (kg)</th> <th rowspan="2">Total availability</th> </tr> <tr> <th>A</th> <th>B</th> <th>C</th> </tr> </thead> <tbody> <tr> <td>P</td> <td>6</td> <td>5</td> <td>2</td> <td>5000</td> </tr> <tr> <td>Q</td> <td>4</td> <td>7</td> <td>3</td> <td>6000</td> </tr> </tbody> </table> | Raw material | Requirement for unit of product (kg) | | | Total availability | A | B | C | P | 6 | 5 | 2 | 5000 | Q | 4 | 7 | 3 | 6000 | | | | |
| Raw material | | Requirement for unit of product (kg) | | | | Total availability | | | | | | | | | | | | | | | | |
| | A | B | C | | | | | | | | | | | | | | | | | | | |
| P | 6 | 5 | 2 | 5000 | | | | | | | | | | | | | | | | | | |
| Q | 4 | 7 | 3 | 6000 | | | | | | | | | | | | | | | | | | |
| Formulate the problem as LPP. | 8 | L3 | 1 | 1 | | | | | | | | | | | | | | | | | | |
| b) Solve the following LPP using Big M method
Max $Z=-2X_1-X_2$ with constraints
$3X_1+X_2=3$
$4X_1+3X_2 \geq 6$
$X_1+2X_2 \leq 4$ $X_1, X_2 \geq 0$ | 12 | L3 | 2 | 1 | | | | | | | | | | | | | | | | | | |

Unit – II

4. a) Find the least cost route for the sales person using Assignment problem.

	A	B	C	D	E
A	---	4	10	14	2
B	12	---	6	10	4
C	16	14	---	8	14
D	24	8	12	---	10
E	2	6	4	16	---

10 L3 4 1

- b) Give the mathematical model for Transportation problem. Find the initial basic solution for the following problem using NWCR

	D1	D2	D3	D4	Supply
O1	11	13	17	14	250
O2	16	18	14	10	300
O3	21	24	13	10	400
Demand	200	225	275	250	

10 L3 3 1

5. a) Solve the following Transportation problem by finding the initial solution using Least cost method.

	D1	D2	D3	D4	D5	Supply
O1	73	40	9	79	20	8
O2	62	93	96	8	13	7
O3	96	65	80	50	65	9
O4	57	58	29	12	87	3
O5	56	23	87	18	12	5
Demand	6	8	10	4	4	

12 L4 3 1

- b) A company is faced with the problem of assigning 4 machines to 6 different jobs. The profits are estimated as follows. Solve the problem to maximize the profit.

	A	B	C	D
1	3	6	2	6
2	7	1	4	4
3	3	8	5	8
4	6	4	3	7
5	5	2	4	3
6	5	7	6	4

8 L3 4 1

6. a) A company has 5 jobs to be done on 5 machines. The cost of doing the jobs in different machines is given. Assign the jobs for different machines to minimize the total cost.

	A	B	C	D	E
1	13	8	16	18	19
2	9	15	24	9	12
3	12	9	4	4	4
4	6	12	10	8	13
5	15	17	18	12	20

8 L3 4 1

- b) Solve the following Transportation problem using uv method. Use VAM to find the initial solution.

	D1	D2	D3	D4	Supply
O1	6	1	9	3	70
O2	11	5	2	8	55
O3	10	12	4	7	70
Demand	85	35	50	45	

12 L4 3 1

Unit – III

7. a) Construct the network for the following data.

Activity	Immediate Predecessor
A	----
B	----
C	A,B
D	B
E	B
F	A,B
G	F,D
H	F,D
I	C,G

6 L3 5 1

- b) A small project is composed of 7 activities whose time estimate are listed in the table:

	Optimistic	Most likely	Pessimistic
1-2	1	1	7
1-3	1	4	7
2-4	2	2	8
2-5	1	1	1
4-6	2	5	8
5-6	3	6	15
3-5	2	5	14

i) Draw the project network.

ii) Find the expected duration and variance of each activity.

iii) Calculate the variance and Standard deviation of project length.

iv) What is the probability that the project will be completed 4 weeks earlier than expected, not more than 4 weeks later than expected. If the project due date is 19 weeks, what is the probability of meeting the due date?

14 L4 5 1

8. a) Discuss the following.

i) Optimistic time estimate ii) Most likely time estimate

iii) Pessimistic time estimate iv) Events and Activity

6 L2 5 1

- b) A project schedule has the following characteristics. From the information,

i) Construct network diagram

ii) Compute earliest time and latest time for each event

iii) Determine the critical path and total project duration

iv) Compute total float and free float for each activity

Activity	Time
1-2	4
1-3	1
2-4	1
3-4	1
3-5	6
4-9	5
5-6	4
5-7	8
6-8	1
7-8	2
8-10	5

14 L4 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
